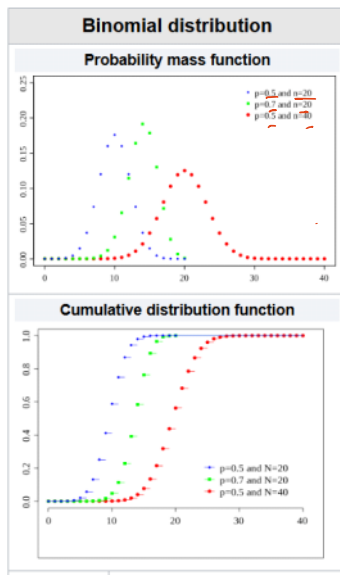




The **binomial distribution** with parameters n and p is the **discrete probability distribution** of the number of successes in a sequence of n **independent experiments**, each asking a **yes-no question**, and each with its own **Boolean-valued outcome**: **success** (with probability p) or **failure** (with probability $q = 1 - p$).

⊗ Discrete Random Variable

⊙ Every outcome of the Experiment is binary

⊙ These experiments are performed for n trials

Ex :- Tossing a coin for 10 Times $n = 10$

Notation: $B(n, p)$. $\{H, T\}$
 $[0, 1]$
 $\downarrow$
 $su \rightarrow$ Success
 $fa \rightarrow$ Failure
 $n \in \{0, 1, 2, \dots, n\}$ no. of trials
 $p \in [0, 1] \rightarrow$ Success
 $q = 1 - p \rightarrow$ Failure

$k \in \{0, 1, 2, \dots, n\} \Rightarrow$ No. of Success $n \Rightarrow$ trials $k \Rightarrow$ $\boxed{nCk}$

$$P_X(k, n, p) = nCk \cdot p^k \cdot (1-p)^{n-k}$$

Binomial coefficient

$$nCk = \frac{n!}{k!(n-k)!}$$

$$\text{mean} = \boxed{n \cdot p}$$

$$\text{Variance} = p - pq$$

$$\text{mean} = p$$

$$\sigma = \sqrt{npq}$$

Ex :- Coin flip

No. of trial (n) = 5

$P(su) = p$ $p = 0.5$

No. of su (k) = 0 to 5

(*) What is the probability of getting exactly 3 heads in a flip

$$n=5$$

$$k=3$$

$n \rightarrow$ total 3 heads

$$P(X=k) = \binom{5}{3} (0.5)^3 (1-0.5)^{5-3} = 0.3125$$

$$= 0.3125$$